

The empirical recurrence for OEIS A302524: recovering a conjecture that is not written down

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Abstract

OEIS A302524 records “Empirical recurrence of order 72”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 72$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 72, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 72 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A302524 is “Number of $n \times 4$ 0..1 arrays with every element equal to 1, 2, 4 or 5 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

$$2, 10, 42, 168, 902, 3256, 15852, 77904, \dots$$

The entry states:

Empirical recurrence of order 72 (see link above)

As of the “Last modified” line on the live entry (Apr 09 2018 09:21:50, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 72$ of the 256, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 144$ exact terms of the model returns order 72 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{72} c_i a(n-i),$$

c_1	4	c_{19}	−10230164	c_{37}	−133329352	c_{55}	−1298858
c_2	3	c_{20}	2938663	c_{38}	7132140	c_{56}	963157
c_3	74	c_{21}	899536	c_{39}	53557652	c_{57}	−198054
c_4	−318	c_{22}	7797742	c_{40}	64593314	c_{58}	−141003
c_5	−166	c_{23}	−848612	c_{41}	35655460	c_{59}	−26542
c_6	−1897	c_{24}	−545447	c_{42}	−125430787	c_{60}	4645
c_7	8636	c_{25}	14184712	c_{43}	128705906	c_{61}	52748
c_8	4839	c_{26}	−5748054	c_{44}	−28863519	c_{62}	−38865
c_9	18690	c_{27}	−10662720	c_{45}	−64639342	c_{63}	9142
c_{10}	−102332	c_{28}	−33768731	c_{46}	14102625	c_{64}	−5779
c_{11}	−76644	c_{29}	39238136	c_{47}	−2593864	c_{65}	7194
c_{12}	−7741	c_{30}	−34153768	c_{48}	14203561	c_{66}	−1058
c_{13}	442482	c_{31}	107962212	c_{49}	−8322898	c_{67}	−68
c_{14}	625004	c_{32}	−45662381	c_{50}	883837	c_{68}	−372
c_{15}	−962554	c_{33}	−3984360	c_{51}	−272714	c_{69}	90
c_{16}	886114	c_{34}	−43003212	c_{52}	7446424	c_{70}	−4
c_{17}	−2354572	c_{35}	−31402472	c_{53}	−5225892	c_{71}	6
c_{18}	4214590	c_{36}	99569221	c_{54}	1734223	c_{72}	−1

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 72, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $72 + 4$ terms removed returns order 72 as well, so $\deg q_{\min} = 72 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $72 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 72 that this sequence satisfies — and that family turns out to have exactly one member.

References

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